

Problem Statement

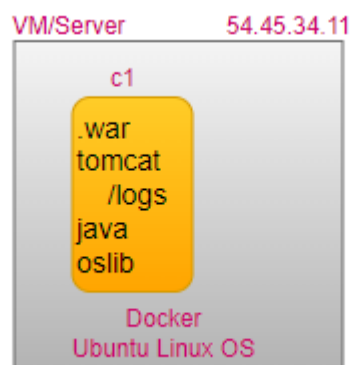
UC: What preventive action to take to save data in case of laptop crash?

take backup outside the laptop

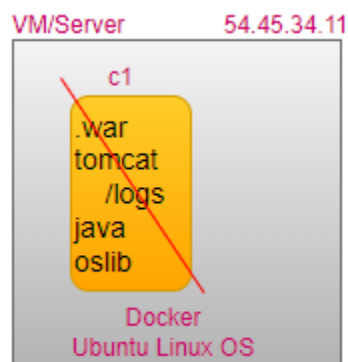
(External storage), external system, either your VM or may be external storage.

Why logs? identify the issue

Running process generates some data e.g., logs, stores with in container



UC: If we lose the container, we lose the data



It is not a good practice to write data inside a container

UC: If there is any issue in your application, where you troubleshoot issue?

logs, **where are these logs?** Inside the container.

If I remove the container, can I get log back?

No, whatever data inside the container will be lost.

`docker container ls` # list of all running containers

`docker container ls -a` # list of all containers

`docker container ls -aq` # list of all stopped container

`docker ps -aq --no-trunc -f status=exited | xargs docker rm`

`docker rm -f 'docker ps -aq'` # delete all containers

`docker ps`

`docker run -d tomcat` # default is latest tag

`docker build -f Dockerfile --tag sampleapp:v2 /root` # in which directory
Dockerfile available

`docker run -d -P sampleapp:v3`

`docker exec -it <CONTAINERID> /bin/bash`

`cd /opt/tomcat`

`ls -l`

`cd logs/`

`ls -l`

`exit`

`docker rm -f <CONTAINERID>` # can we get the log /recover the data 😞

Solution: docker volume

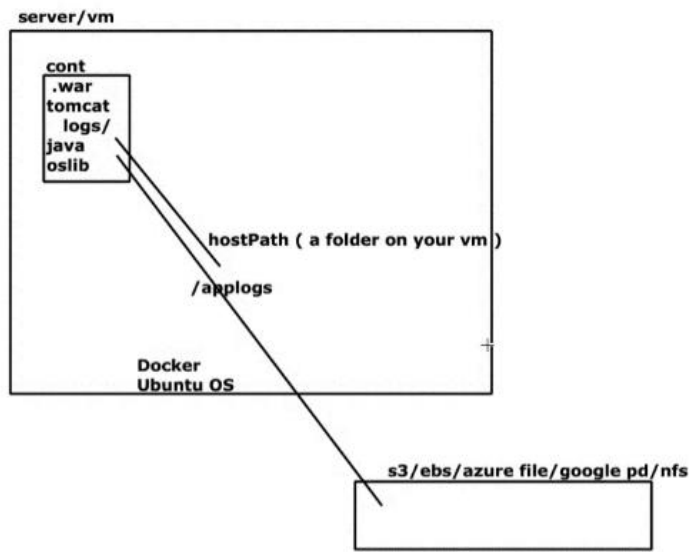
Rather than writing data inside the container, write the data into **NAS/SAN**

NAS: Network attached Storage

SAN: Storage Area Network

Docker Volume just like external storage, helps to make data persistent

Volume (Storage Drive)-**persistent outside the container**



docker volume ls

docker volume create applogs

docker volume inspect # /var/lib/docker/volumes/applogs/-data

cd /var/lib/docker/volumes/applogs/-data

ls-l # nothing is here

We can't attach a volume to **existing running container**, attach volume while creating a container

docker build -f myapp --tag myapp:v2 /root # in which directory Dockerfile available

docker run -d -P --mount type=volume,src=applogs,target=/opt/tomcat/logs myapp:v1

Binds the **container path with target** with **local host path with src**

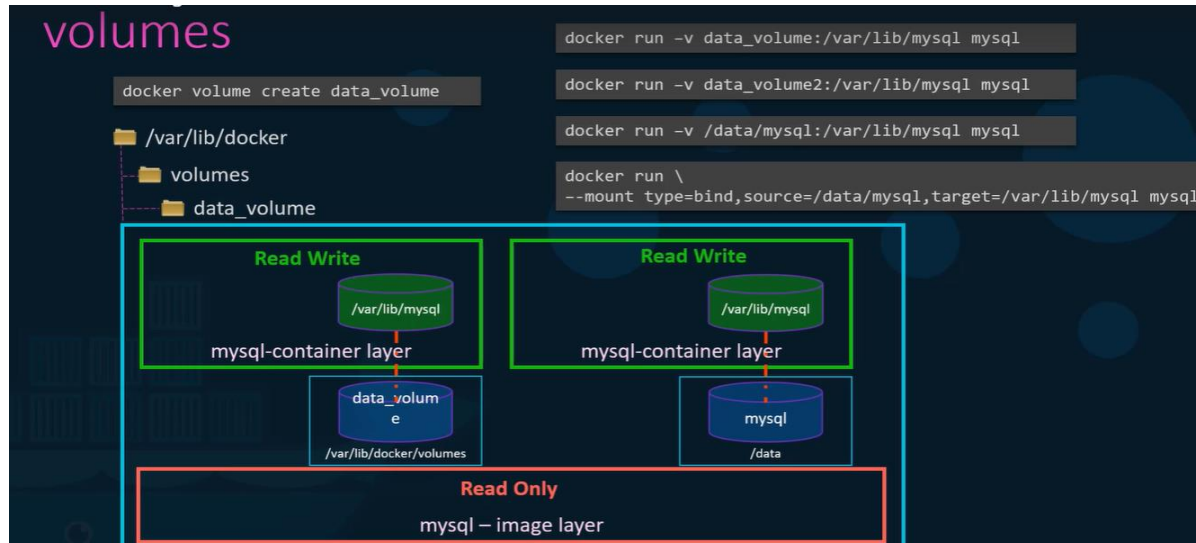
docker ps

cd /var/lib/docker/volumes/applogs/-data

ls-l # data is persistent

`docker rm -f <CONTAINERID>`

can we get the log /recover the data 😊



Concept of docker volume is to keep the data outside the container.

`docker rm -f 'docker ps -qa'` # remove all containers

step1: where are the container logs available?

UC1: Create Container **without** volumes

`docker run -d -P sampleapp: v3`

`docker ps`

`docker exec -it <container id> bash`

`cd /opt/tomcat/logs`

`ls -l`

`ctrl pq`

`docker rm -f <container id>`

```

root@node1:~# docker image ls
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
lerndevelops/myapp   sampleapp-v3    1fa9c81f4ee3   45 minutes ago 555MB
sampleapp            v3              1fa9c81f4ee3   45 minutes ago 555MB
sampleapp            v2              13e5fc24be3b   50 minutes ago 555MB
sampleapp            v1              04ea872b24a4   53 minutes ago 555MB
myapp                v2              7dfb7e44e218   About an hour ago 649MB
myapp                v1              37c654d807c5   About an hour ago 649MB
tomcat               latest          345867df0879   9 days ago     649MB
ubuntu               18.04          c090eaba6b94   9 days ago     63.3MB
nginx                latest          f6d0b4767a6c   2 weeks ago    133MB

root@node1:~# docker run -d -P sampleapp:v3
b2e2046e2532ae018f7be1451869793f461e322d894972e9dcd4eac010ab2144

root@node1:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
b2e2046e2532   sampleapp:v3   "/bin/sh -c '/opt/to..." 7 seconds ago  Up 6 seconds  0.0.0.0:49156->8080/tcp    nice_mestorf

root@node1:~# docker exec -it b2e2046e2532 bash
root@b2e2046e2532:~# cd /opt/tomcat/
root@b2e2046e2532:/opt/tomcat# ls -l
total 144
-rw-r----- 1 root root 19318 May  5 2020 BUILDING.txt
-rw-r----- 1 root root  5408 May  5 2020 CONTRIBUTING.md
-rw-r----- 1 root root  57011 May  5 2020 LICENSE
-rw-r----- 1 root root  1726 May  5 2020 NOTICE
-rw-r----- 1 root root  3255 May  5 2020 README.md
-rw-r----- 1 root root  7136 May  5 2020 RELEASE-NOTES
-rw-r----- 1 root root 16262 May  5 2020 RUNNING.txt
drwxr-x--- 2 root root  4096 Jan 30 15:49 bin
drwxr-x--- 1 root root  4096 Jan 30 16:42 conf
drwxr-x--- 2 root root  4096 Jan 30 15:48 lib
drwxr-x--- 1 root root  4096 Jan 30 16:42 logs

```

log file is inside the container, if we remove the container, logs will also lost.

UC2: Create Container **with** volumes

docker volume ls

docker volume create myvol # local volume

docker volume ls

docker volume inspect myvol # /var/lib/docker/volumes/myvol/_data

docker run -d -P --volume myvol:/opt/tomcat/logs sampleapp: v3

cd /var/lib/docker/volumes/myvol/_data

ls -l

For existing container, we can't attach a volume, volume can be attached during container creation time.

UC3: custom location (bind mount)

mkdir nginxconfig

mkdir nginxlogs

docker run -d -P --mount

type=bind,src=/root/nginxconf,target=/usr/share/nginx/html --mount

type=bind,src=/root/nginxlogs,target=/var/log/nginx nginx:latest

```
docker volume ls
```

```
docker volume create myvol          # local volume
```

```
docker volume ls
```

```
docker volume inspect <Id>
```

```
cd /var/lib/docker/volumes/myvol/_data
```

```
ls -l                                # no data
```

```
docker run -d -P --volume myvol:/opt/tomcat/logs sampleapp:v3  # local host:  
<container path>
```

```
cd /var/lib/docker/volumes/myvol/_data
```

```
touch neeraj.txt
```

```
docker exec -it <containerid> bash
```

```
cd /opt/tomcat/logs
```

```
ls -l
```

```
touch vohra.txt
```

```
exit
```

```
cd /var/lib/docker/volumes/myvol/_data
```

```
ls -l
```

customized path: bind mount

```
mkdir /logs
```

```
mkdir /config
```

```
docker run -d -P --volume /logs:/var/log/nginx --volume /config:  
/usr/share/nginx/html nginx
```

```
cd /logs
```

```
ls -l
```

```
cd /config
```

```
ls-l
```

```
touch index.html
```

```
echo "My Nginx Container" > index.html
```

```
ls -l
```

```
cd /logs
```

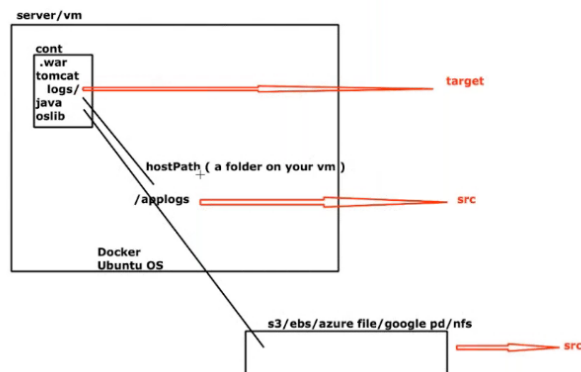
```
tail -f access.log
```

```
docker run -d -P --mount type=bind,source=/config,target=/usr/share/nginx/html  
nginx #dst
```

```
docker run -d -P --mount  
type=volume,source=myvol,target=/usr/share/nginx/html nginx #dst
```

docker volumes
helps to make the data persistent

```
docker run -d -P --mount type=bind,src=/root/nginxconf,target=/usr/share/nginx/html --mount  
type=bind,src=/root/nginxlogs,target=/var/log/nginx nginx:latest
```



What if data is in different server?

We have to keep data in central location like S3.

